DUC-QUANG (CILLIAN) NGUYEN

AI Engineer | Backend & Machine Learning Engineer

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PROFESSIONAL SUMMARY

AI and backend engineer with 3 years at MB Securities (MBS), a top-five Vietnamese securities firm by market share of trading value and a subsidiary of MB Bank.

2.5M questions answered • **~50,000** users per month • **~AUD 230M** transactions referenced • **3×** Sao Khue Award

- Led the AI architecture behind **DolphinAI**, a production multi-agent LLM system in the MBS mobile app and web platform.
- Set the AI integration patterns for the MBS **on-premise CRM AI platform** on LangGraph, LangChain and MCP.
- The AI specialist inside **cross-functional teams of 15 to 30+ people**, turning business requirements into systems that hold up in production and deliver commercial results.
- Now based in Sydney, starting a **Master of Computer Science** at the University of Sydney.
- Seeking **AI Engineer**, **Forward Deployed Engineer**, **AI Consultant** or **Backend Software Engineer** roles, or similar hands-on, customer-facing AI positions.

TECHNICAL SKILLS

Languages: Python (OOP, asyncio, multithreading), C, SQL, Bash

AI & LLM Systems: Multi-agent orchestration, LangGraph, LangChain, Model Context Protocol (MCP) / FastMCP server design, RAG, agent framework design, prompt engineering, Langfuse tracing and evaluation, embedding pipelines, Hugging Face, Text-Embeddings-Inference, Qdrant

ML & Data: Predictive modelling (churn, propensity, lead scoring), feature engineering, MLflow, Airflow, model serving; PostgreSQL, Redis, SQL Server, Google BigQuery, Cloud Firestore, dbt, data modelling

Backend & APIs: FastAPI, gRPC, RESTful API design, Server-Sent Events (SSE) streaming, Envoy Proxy, authentication, rate limiting, caching, unit testing, JMeter load testing, horizontal scaling

Infrastructure & Tooling: Linux (AlmaLinux), Git, on-premise deployment, cron scheduling, n8n; Scrapy, Selenium, BeautifulSoup and distributed crawler design; Claude Code, OpenAI Codex and Cursor for agentic workflows

PROFESSIONAL EXPERIENCE

MB Securities (MBS)

Hanoi, Vietnam

AI Engineer | Data Science Specialist

Sep 2024 – Jul 2026

Products: **DolphinAI**, an AI stock-market assistant for retail investors on the MBS mobile app and web platform (launched Feb 2025; 15-person team); and the **CRM AI platform**, on-premise agentic intelligence for the MBS broker network (30+ person programme).

The AI specialist between business stakeholders and engineering on both: scoping which problems were worth solving with AI, and preferring the simplest design that met the requirement.

DolphinAI: AI systems engineering

- Designed and built DolphinAI's in-house **multi-agent system**, and owned the AI core within the product team.
- Grounded answers in **RAG** over MBS market data, on a horizontally scalable service architecture.
- Built a **reusable AI Agent framework** standardising how agents are defined, routed and tested.
- Wrote the full suite of production agents and the routing logic, so new capabilities ship without touching orchestration.
- Grounded **97.2% of responses** in proprietary MBS knowledge.
- Built the distributed crawling pipeline behind it, normalising **13 reputable news sources** continuously.

DolphinAI: backend & platform engineering

- Built the **RESTful and gRPC APIs** behind DolphinAI: authentication, rate limiting, database integration, embedding services, unit tests.
- Added **Server-Sent Events (SSE) streaming** for token-by-token agent responses.
- Scaled high-computation services **from 8 to 3,000+ concurrent users (375×)** using async service design and Envoy Proxy load balancing.
- Verified under JMeter load testing: **99.999% of production queries returned in under 15 seconds**.
- **Cut projected system cost 8×** by building the agent stack in-house instead of licensing an external platform.

CRM AI platform: architecture & engineering enablement

- Set the AI integration approach for the **30+ person CRM programme**.
- Built a **reference LangGraph agent workflow** defining the orchestration, state-management, parallel-execution and model-tiering patterns.
- Designed the connectivity layer between company data and agents: **dedicated MCP servers** over the data warehouse, CRM APIs and PostgreSQL.
- Shipped reusable MCP templates so other applications across MBS use the same tooling.

- Ran the platform **entirely on-premise** for financial data residency, with **Langfuse** for tracing, evaluation and cost observability.

Machine learning: commercial modelling

- Led a **team of three data scientists** on churn, margin-lending propensity and MB Group lead-generation models.
- Built the **FastAPI / MLflow / Airflow** training and serving platform behind them: reproducible runs, one-step deployment.
- Margin-propensity model identified and influenced **~55% of weekly margin users** on digital channels.

Business impact

- **2.5M user questions** answered as of **February 2026**, with **~50,000 users per month**.
- Customers referenced DolphinAI for over **4,300 billion VND (~AUD 230M)** in transactions as at December 2025.
- Contributed to a **10% increase** in MBS's digital-channel market share.
- Work contributed to **three Sao Khue Awards**, one of them a dedicated award for DolphinAI.

MB Securities (MBS) Hanoi, Vietnam

Data Science Intern → Data Science Apprentice Aug 2023 – Sep 2024

- Migrated and streamed data from **Cloud Firestore into Google BigQuery**, preserving data integrity across the move.
- Built a **proof-of-concept customer service chatbot** in n8n, validating conversational automation ahead of DolphinAI.
- Ran cross-sell analysis across the MB Bank and MBS customer bases to find joint-product opportunities.

VNPT-Media Hanoi, Vietnam

Data Science Intern (Part-time) Mar 2023 – Jun 2023

- Supported data analysis and Python data processing workflows at one of Vietnam's largest telecommunications and digital media groups.

EDUCATION

University of Sydney, Master of Computer Science Aug 2026 – Aug 2028 (expected)

National Economics University, Bachelor of Data Science (Mathematical Economics), Distinction Aug 2020 – Aug 2024

AWARDS, CERTIFICATIONS & LANGUAGES

- **Sao Khue Award 2025:** dedicated award for DolphinAI (Vietnam's top IT industry accolade, awarded annually by VINASA); contributed to three Sao Khue-recognised products.
- **Outstanding Employee, MB Group 2025:** one of **21 recipients from 15,000 employees** across MB Bank and its subsidiaries • **High-Potential Employee, MB Securities 2024**
- IELTS 8.0, Jan 2026
- **DeepLearning.AI**, Jul 2023: Neural Networks and Deep Learning; Mathematics for Machine Learning and Data Science Specialization
- Languages: Vietnamese (native), English (full professional proficiency)

WORK RIGHTS & AVAILABILITY

- Based in Mascot, Sydney, with current Australian work rights.
- **48 hours per fortnight** during study periods; **unrestricted during course breaks**; **full-time from August 2028**.
- Referees available on request.